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# Investigation on the impact of powder arching in small die filling

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ABSTRACT

The flow of particulate materials is critical during processes such as mixing, compression and packing. Non-cohesive arching, a feature characteristic of coarse and free-flowing particles, has been studied extensively for silos and hoppers. However, the arching of powders during die fill has received much less attention. In this study, die fill performance of coarse and free-flowing nonpareils was evaluated using a specially designed die filling device in order to investigate the impact of non-cohesive arching during die fill under gravity flow. Through evaluating die fill performance, the arching phenomenon during dynamic conditions of die fill could be captured. Nonpareils with large particle size increased the likelihood of arching and caused poorer die fill performance for narrow orifices. In contrast, die fill in large orifices was generally better with larger particles due to reduced inter-particulate friction. Both particle size and size distribution influenced non-cohesive arching during die fill. Forced feeding did not appear to affect die fill performance and non-cohesive arching. A critical particle size range beyond which die fill performance would decrease, particularly for the narrow orifices, was identified. Findings from this study provided a better insight into non-cohesive arching during die fill.

## 1. Introduction

Materials in granular or powder form are widely utilised in many industries including the pharmaceutical industry. The manufacture of oral tablets typically involves powder transfers, die filling and compaction of the feed material into tablets. Good flow of the feed is essential to ensure reproducibility of the die filling step, which in turn affects critical quality attributes of tablets such as weight uniformity and mechanical strength. Powders with good flowability could achieve higher die fill densities (Freeman and Fu, 2008; Wu et al., 2012) and lower die fill variation (Fassihi and Kanfer, 1986). However, it has also been suggested that powder flow properties are not correlated with die fill performance; poor flowing powders have been reported to achieve more uniform die fill than those with better flow (Mills and Sinka, 2013; Nicklasson and Nyqvist, 1982; Sinka et al., 2004).

Powder flow also affects other handling and processing operations which include discharge from hoppers and silos, blending and conveyance (Fitzpatrick et al., 2004). However, due to the complexity of inter-particle interactions, bulk behaviour dynamics of granular material is still not well understood (Lumay et al., 2012). Development of clogs or arches that obstruct the powder flow in bins or silos by gravity is an important characteristic of granular materials and a common problem encountered during powder handling and discharge after storage (Schulze, 2007; Zuriguel, 2014).

An arch is a structure comprising mutually stabilised particles (Janda et al., 2008; Zuriguel et al., 2005). Arching occurs when the powder particles interlock with each other and form a stable arch over the exit orifice. A narrow exit orifice or passage through a bottleneck can lead to arching for both cohesive and free-flowing powders (Baserinia et al., 2016). Arching entails a complete stop of the powder flow unless external stresses, such as vibrations or tapping, are applied to break the arch. Consequently, powder arching in silos and hoppers can lead to intermittent flow of material or even force stoppage of production lines (Oldal et al., 2012; Zuriguel, 2014). For fine, cohesive powders, the arch forms due to cohesive strength of the bulk solid (Schulze, 2007), which is related to unconfined yield strength, angle of internal friction and flow function of the powder. By determining these parameters, it is possible to estimate the minimum hopper angle and hopper outlet size for mass flow to occur while reducing the likelihood of arching for a given powder (Drescher et al., 1995a,b; Jenike, 1964). For coarse-grained, free-flowing materials, the arch forms due to mechanical interlocking and wedging between the powder particles. Increasing the diameter of the exit orifice typically solves both types of arching since the larger the outlet size, the higher is the likelihood that any arch formed destabilises.

Physical mechanisms underlying the formation of non-cohesive arches have been studied extensively using 2D and 3D silos as well as numerical simulations (Arevalo and Zuriguel, 2016; Garcimartín et al.,

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2009). Distribution of avalanche size (defined as the number of discrete particles that flow from the point of arch destruction until the development of a new, stable arch) is often used to quantify non-cohesive arching in these studies. The probability of non-cohesive arching decreased from 1 to zero when the ratio between exit orifice size and particle diameter increased from 2 to 5 in a 2D silo (To et al., 2001). Similar findings were also reported and the possibility of critical ratio beyond which no arching would occur for spherical particles was also studied (Janda et al., 2008; To, 2005; Zuriguel et al., 2005). Besides particle size, particle shape was also found to have critical impact on arching; grains with anisotropic shapes increase the likelihood of non-cohesive arching (Ashour et al., 2017; Zuriguel et al., 2005).

Similar to powder discharge from hoppers and silos, the act of die filling involves the flow of material from a receptacle, in this case a feed shoe or feed frame depending on the tablet press design, into an orifice. While much research work has focused on powder arching in hoppers and silos, the arching of powders during die fill has received much less attention. Certain pharmaceutical applications, such as production of mini-tablets (tablets with diameters smaller than 6 mm), may face problems with non-cohesive arching during die filling simply due to the narrow die orifices involved (Aleksovski et al., 2015). A study which examined powder flow rates through different orifice diameters recommended a maximum particle size of no more than 1/3 of the die diameter to prevent blockage of the die opening (Flemming and Mielck, 1995). This principle is supported in another study which found that stronger mini-tablets with lower variation in weight and hardness can be produced by removing granules of size fractions larger than 1/3 of the die diameter prior to compression (Zhao et al., 2017). Nonetheless, little literature information is available on non-cohesive arching in relation to die filling. Furthermore, determination of arching events during die filling is more challenging than that for hoppers and silos since the system is dynamic; powder is being moved across and into the orifice concurrently during the die fill process. In most modern rotary tablet presses, the tablet feed is circulated continuously and fed into dies by rotating paddles in a feed frame. Different designs of the paddle are available to change the circulation and forced feeding mechanism (Mendez et al., 2010). The purpose of this work was two-fold: to develop a methodology of evaluating non-cohesive arching during die filling and study how arching could potentially affect die filling of coarse and free-flowing materials under different processing conditions such as orifice diameter and forced feeding.

## 2. Materials and methods

### 2.1. Materials

Nonpareils (sugar spheres, Pharm-a-spheres®, Hanns G. Werner, Germany) of various size fractions (250–300 µm, 355–425 µm, 500–600 µm, 600–710 µm, 710–850 µm and 850–1000 µm) were used as model coarse and free-flowing materials.

### 2.2. Preparation of nonpareil blends

Binary blends of nonpareils from the various size fractions were prepared to obtain blends with different particle size and size distribution. Two sets of binary blends were prepared, Set A: 13 blends from neighbouring size fractions; Set B: 20 blends from 250 to 300 µm size fraction and one of the remaining size fractions other than 355–425 µm (Table 1). Blends were prepared by mixing the nonpareils in a double cone mixer (AR401, Erweka, Germany) rotated at 15 rpm for 5 min. A total of 33 nonpareil blends were thus prepared in this study.

### 2.3. Characterisation of particle size and shape

Random samples of nonpareil blends were obtained using a sample divider (PT100, Retsch, Germany). Particle size and size distribution of the samples were determined using a laser diffractometer (LS230, Beckman Coulter, US) fitted with the dry powder module. Target obscuration level was set at about 5–7%. A cumulative undersize curve was generated from which particle size parameters corresponding to the 10th, 50th and 90th percentiles ($D_{10}$, $D_{50}$ and $D_{90}$) were determined. Particle size distribution was reflected by Span, calculated using Eq. (1). Three readings were taken for each nonpareil blend and the results averaged.

$$\text{Span} = \frac{D_{90} - D_{10}}{D_{50}} \quad (1)$$

Images of 600 nonpareil particles from the original size fractions (250–300 µm, 355–425 µm, 500–600 µm, 600–710 µm, 710–850 µm and 850–1000 µm) were taken using a stereomicroscope (SZ61, Olympus, Japan) connected to a video camera (DP71, Olympus, Japan). Imaging software (Image Pro Plus 6.3, Media Cybernetics, US) was used for particle shape analyses. Aspect ratio (AR) of the particles was calculated using Eq. (2). A larger aspect ratio would indicate that the particle is more elongated.

$$\text{AR} = \frac{\text{Major axis}}{\text{Minor axis}} \quad (2)$$

where the major and minor axes are the largest and smallest diameters of an ellipse with an area equivalent to that of the particle, respectively.

Particle roundness was determined using Eq. (3). A value of 1 would indicate that the particle is a perfect circle while values greater than

**Table 1**
Composition of binary nonpareil blends of Sets A and B.

| Size fraction F1 (µm) | Size fraction F2 (µm) | F1 (%, w/w) | F2 (%, w/w) | Binary blend code |
|---|---|---|---|---|
| *Set A* | | | | |
| 250–300 | 355–425 | 100 | 0 | A1 |
| | | 70 | 30 | A2 |
| | | 50 | 50 | A3 |
| | | 30 | 70 | A4 |
| 355–425 | 500–600 | 100 | 0 | A5 |
| | | 70 | 30 | A6 |
| | | 50 | 50 | A7 |
| | | 30 | 70 | A8 |
| 500–600 | 600–710 | 100 | 0 | A9 |
| | | 70 | 30 | A10 |
| | | 50 | 50 | A11 |
| | | 30 | 70 | A12 |
| 600–710 | – | 100 | 0 | A13 |
| *Set B* | | | | |
| 250–300 | 850–1000 | 10 | 90 | B1 |
| | | 30 | 70 | B2 |
| | | 50 | 50 | B3 |
| | | 70 | 30 | B4 |
| | | 90 | 10 | B5 |
| 250–300 | 710–850 | 10 | 90 | B6 |
| | | 30 | 70 | B7 |
| | | 50 | 50 | B8 |
| | | 70 | 30 | B9 |
| | | 90 | 10 | B10 |
| 250–300 | 600–710 | 10 | 90 | B11 |
| | | 30 | 70 | B12 |
| | | 50 | 50 | B13 |
| | | 70 | 30 | B14 |
| | | 90 | 10 | B15 |
| 250–300 | 500–600 | 10 | 90 | B16 |
| | | 30 | 70 | B17 |
| | | 50 | 50 | B18 |
| | | 70 | 30 | B19 |
| | | 90 | 10 | B20 |

unity would indicate deviation from a circle.

$$\text{Roundness} = \frac{P^2}{4\pi A} \quad (3)$$

where P is the perimeter and A is the projected area of the particle image.

### 2.4. Flow characterisation of nonpareils

#### 2.4.1. Determination of angle of repose

Approximately 100 g of sample from each nonpareil blend was passed through a funnel onto a circular base plate (Copley Scientific, United Kingdom) so that it formed a conical heap. Height of the funnel was adjusted such that the orifice was about 2–4 cm away from the powder heap. Height of the cone was measured and used to calculate angle of repose (AoR) using Eq. (4). The procedure was repeated three times and the results averaged.

$$\text{AoR}(^\circ) = \tan^{-1}\left(\frac{h}{r}\right) \quad (4)$$

where h is the height of the heap formed, r is the radius of the base.

#### 2.4.2. Compressibility test

Nonpareil samples were added to a pre-weighed 100 mL graduated cylinder. After scraping excess particles carefully from the top of the cylinder, the cylinder was tapped at 250 drops/min using a jolting volumeter (J. Engelsmann AG, Germany) until a constant tapped volume was obtained. Bulk density ($\rho_{bulk}$) and tapped density ($\rho_{tapped}$) were calculated using Eqs. (5) and (6), respectively.

$$\rho_{bulk}\ (\text{g. cm}^{-3}) = \frac{\text{Sample mass}}{100\ \text{mL}} \quad (5)$$

$$\rho_{tapped}\ (\text{g. cm}^{-3}) = \frac{\text{Sample mass}}{\text{Tapped volume}} \quad (6)$$

$\rho_{bulk}$ and $\rho_{tapped}$ were used to calculate Hausner ratio (HR) using Eq. (7). The procedure was repeated three times for each blend and the results averaged.

$$\text{HR} = \frac{\rho_{tapped}}{\rho_{bulk}} \quad (7)$$

#### 2.4.3. Powder rheometry

A FT4 powder rheometer (FT 4, Freeman Technology, United Kingdom) was used to measure the basic flow energy (BFE) and specific energy (SE) of the nonpareils. Samples were added to a 50 mm diameter test cylinder. Due to the relatively large particle size of the nonpareils, a twisted blade with diameter of 23.5 mm was used for the test in order to allow for sufficient clearance between the blade and the cylinder wall. The twisted blade moved in a series of downward clockwise and upward anti-clockwise rotations through the nonpareils while measuring the torque resistance. BFE was the amount of energy required to displace the particles in the test cylinder during the downward traverse under high stress flow conditions. SE was measured during the upward traverse, where the nonpareils flowed under low stress conditions. The test protocol and related flow parameters have been reported in greater detail elsewhere (Freeman, 2007; Leturia et al., 2014; Zhou et al., 2010). The procedure was repeated at least two times for each blend and the results averaged.

### 2.5. Die fill experiments

A die filling device was used to carry out the die fill experiments in this work. Three-dimensional models of the device were designed using software (Solidworks® 2016, Dassault Systems, US) prior to fabrication. The device comprised a mounted aluminium platform, feeder and collection cups. Four equidistant, circular "die orifices" of different diameters (2, 4, 7 and 10 mm) were drilled through the entire thickness of the platform (Fig. 1A). The die orifices were 20 mm deep and volume of each cup was approximately 24910 $mm^3$. One collection cup was mounted below each orifice on the underside of the platform. The feeder unit consisted of a metallic frame with four swappable paddle inserts (Fig. 1B), each aligned to one orifice when the feeder was in position on the platform. In order to simulate different effects of forced feeding on feed frames of rotary tablet presses, paddle inserts with different angles of inclination (30, 60 and 90°) were also available. Operation of the die filling device was as follows. For each die orifice, nonpareil samples were first filled into the corresponding trench of the feeder unit (Fig. 1C). Excess particulates were carefully scraped off from the feeder to ensure consistent material head height. A rotor, operating at a linear velocity of 4 m/min, was attached to the feeder unit via a metal cord and actuated its movement across the platform. The linear velocity was chosen to be close to reported feed paddle speeds on rotary tablet presses (Goh et al., 2017; Peeters et al., 2015, 2016) and give sufficient time for die fill such that particle size, rather than paddle speed, would likely be the major contributor to non-cohesive arching in this study. Initial position of the feeder on the platform was kept

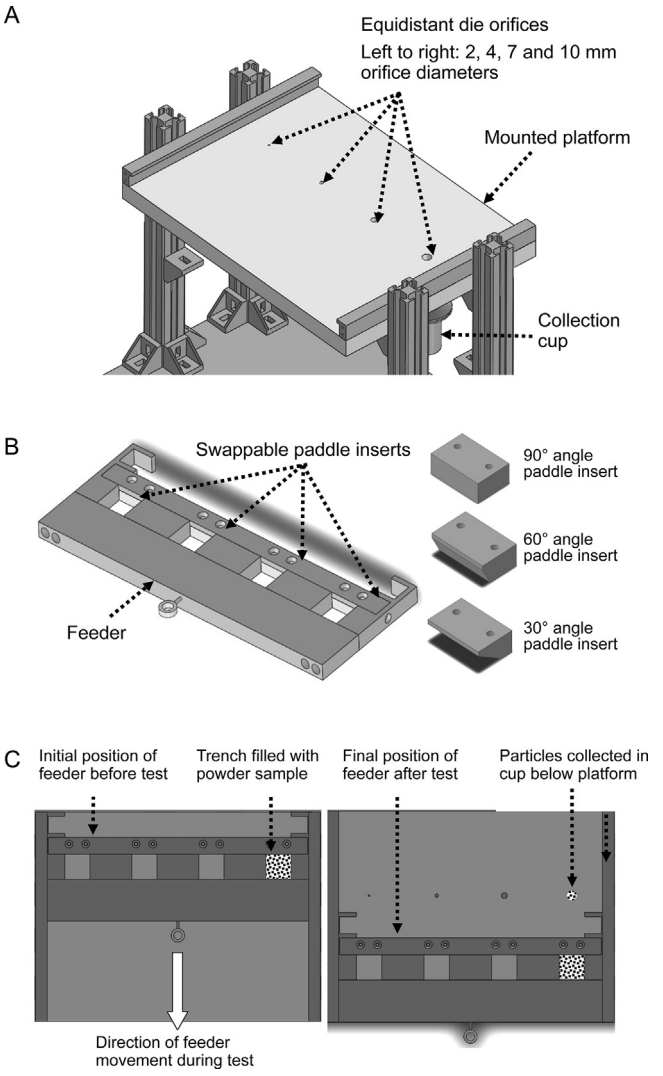



**Fig. 1.** (A) Mounted platform of die fill device illustrating the four die orifices with one collection cup below each orifice. (B) Feeder with swappable paddle inserts of different inclination angles. (C) Trench in feeder was filled with sample to same head height. Sample entered the collection cup via the orifice as the feeder moved across the platform.

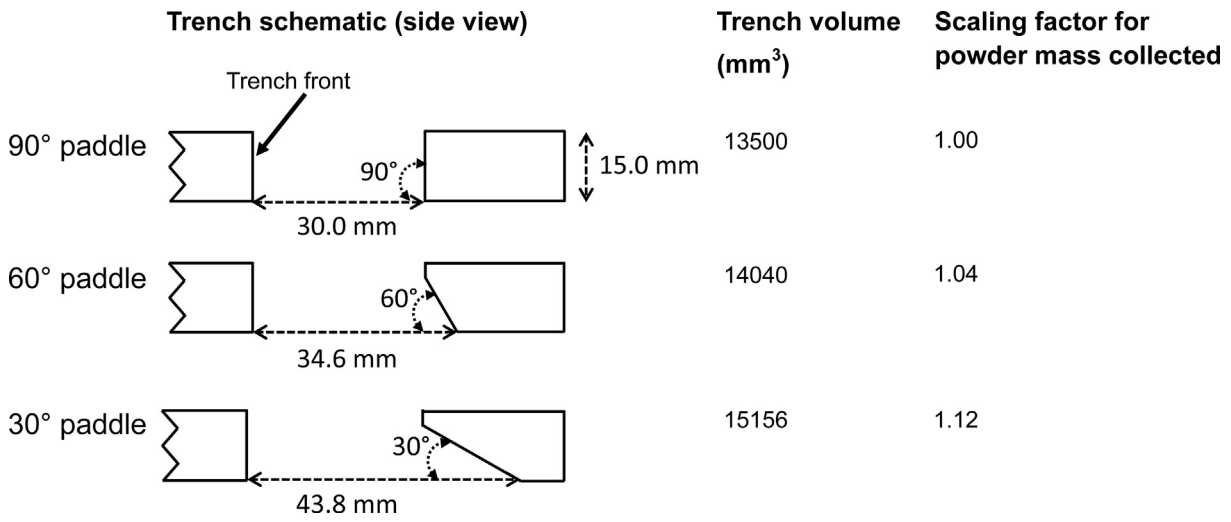



**Fig. 2.** Trench schematics, calculated trench volumes and scaling factors for powder mass collected across different paddle angles.

constant to ensure consistent time for acceleration between different measurements. As the feeder moved across the platform, nonpareils in the trench were also pushed by the paddle insert across the die orifice with some particles entering the collection cup below the platform by gravitational effect. Mass of nonpareils collected in the cup was measured.

The trench volume was larger for paddle inserts with smaller angles (Fig. 2). Measured values of powder masses collected in the cups for the smaller paddle angles were thus scaled using the ratio between the respective trench volume and that of the 90° paddle in order to account for the initial higher amount of powder mass. The scaling factor was calculated as shown in Eq. (8).

$$\text{Scaling factor} = \frac{\text{Trench volume of paddle}}{\text{Trench volume of } 90^{o} \text{ paddle}} \quad (8)$$

The distance between the bottom part of the paddle and trench front was also greater for the smaller paddle angles, which implied longer time of contact between the powder sample and the orifice. Mass flow rate of the powder was calculated using Eq. (9) taking into consideration the time taken for the trench to completely pass across the die orifice and the scaling factor.

$$\text{Mass flow rate (g. s}^{-1}) = \frac{\text{Mass of collected nonpareils}}{\text{Time}} \div \text{Scaling Factor} \quad (9)$$

Normalising mass flow rate against orifice cross-sectional area would enable comparisons to be made across the different orifice diameters (#), for # = 10, 7, 4 or 2 mm. Mass flow rate per unit orifice area (MFA) was calculated using Eq. (10). Higher MFA values were indicative of better die fill performance.

$$\text{MFA}_{\#}(\text{g. cm}^{-2}\text{. s}^{-1}) = \frac{\text{Mass flow rate}}{\text{Orifice cross-sectional area}} \quad (10)$$

Die filling experiments were done for all of die orifice diameters from 2 to 10 mm using nonpareil blends from Set A and paddle insert with 90° angle of inclination. For the 2 mm die orifice, additional experiments were carried out using nonpareil blends from Set B and paddle inserts with 30 and 60° angles of inclination. At least three replicate measurements were taken for each experimental configuration and the results averaged. All die fill experiments were conducted under controlled ambient temperature and relative humidity conditions (25 °C, 50%).

### 2.6. Assessment of measurement reproducibility

In order to assess the reproducibility of measurements obtained from the die filling device, mean and standard deviation of the nonpareil mass were used to calculate relative standard deviation (RSD) using Eq. (11) for each factor combination.

$$\text{RSD}(\%) = \frac{\text{Standard deviation of powder mass}}{\text{Mean powder mass}} \text{x}100\% \quad (11)$$

Average RSD value for each die orifice was reported.

### 2.7. Classification of orifice flow behaviour under static conditions

For each nonpareil blend from Sets A and B, visual observations of the orifice flow under static conditions were also made. This was carried out by first positioning the trenches directly above the die orifices. A thin piece of plastic film, with its end attached to a retracting line, was used to cover the die orifices before nonpareils were added into the trench. A rotor then retracted the film quickly at a linear speed of 12 m/min. Flow behaviour of the nonpareils into the orifices was noted. When the particles were able to enter the orifice without any evidence of arching or blockage, the flow was classified as “Type 0” (Fig. 3A). In cases where the flow was arrested by arching, a waiting time of five seconds was given to ensure that the arch was stable. If arching was observed after some outflow of particles through the orifice, it was classified as a “Type 1” flow (Fig. 3B). On the other hand, if the flow was arrested immediately after the film was removed, the arch would be classified as a “Type 2” flow (Fig. 3C). For each experimental configuration, five replicated observations were made and the orifice flow behaviour was characterised by the highest level of classification (Type 2 being the highest and Type 0 the lowest) obtained among the observations.

## 3. Results

### 3.1. Physical characteristics of nonpareil blends

Particle shape data of the original nonpareil size fractions are as follows: 250–300 μm (AR: 1.20 ± 0.15, Roundness: 1.19 ± 0.15), 355–425 μm (AR: 1.16 ± 0.16, Roundness: 1.26 ± 0.31), 500–600 μm (AR: 1.09 ± 0.07, Roundness: 1.13 ± 0.02), 600–710 μm (AR: 1.13 ± 0.11, Roundness: 1.16 ± 0.14), 710–850 μm (AR: 1.12 ± 0.11, Roundness: 1.13 ± 0.04) and 850–1000 μm (AR: 1.08 ± 0.07, Roundness: 1.12 ± 0.02). Nonpareils were thus relatively round in shape and did not differ significantly from one another

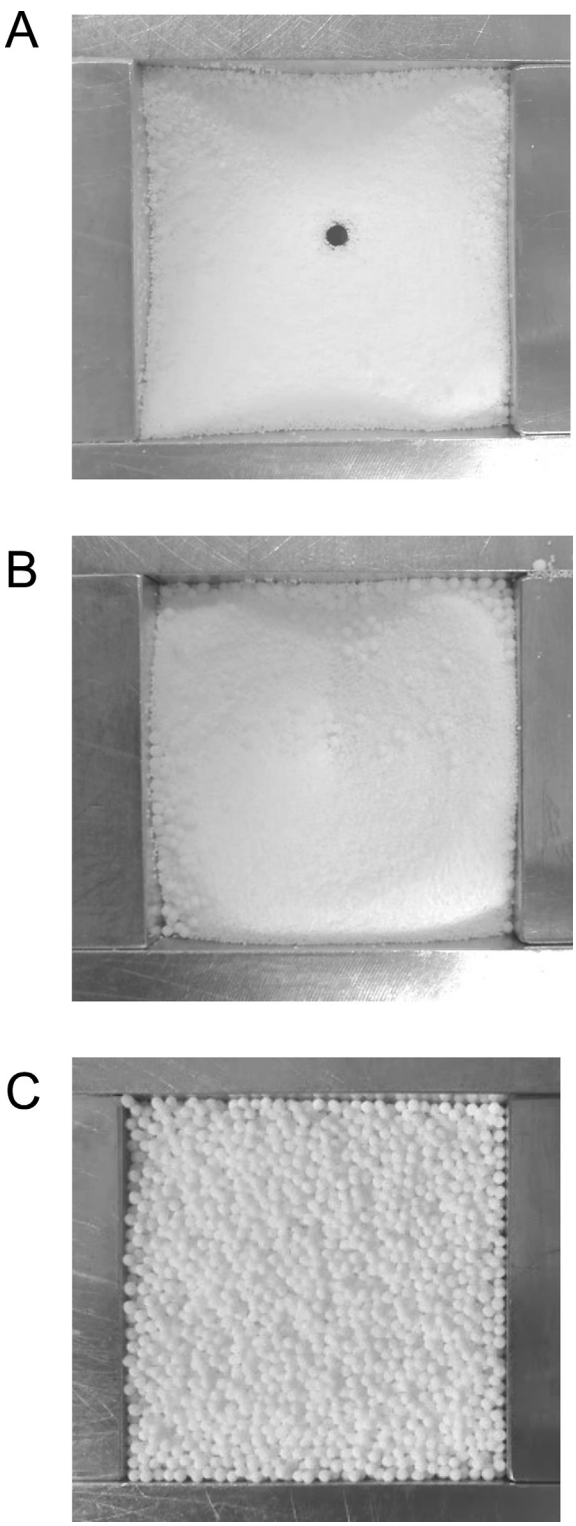



**Fig. 3.** Types of observed flow behaviour under static conditions after plastic film above orifice is removed: (A) Type 0 – no arching observed, (B) Type 1 – arching observed after some outflow of material through orifice and (C) Type 2 – arching observed immediately after film is removed.

in terms of elongation and roundness. This also implied that the particle shape of the binary blends did not differ significantly and was thus likely not a confounding factor for the arching analyses in this study.

Table 2 shows the particle size distribution and bulk densities of the nonpareil blends. While the blends had a wide range of $D_{10}$, $D_{50}$ and $D_{90}$ values, bulk densities had a relatively narrow range between 0.82 and 0.94 g/cm$^3$. This could be attributed to the similar bulk densities of the original size fractions used to prepare the binary blends. It could also be seen that Set A blends generally had lower Span values and narrower particle size distribution than Set B blends, which could be due to their difference in compositions: Set A blends were prepared from neighbouring size fractions while size fractions used to prepare Set B blends were more different in particle size. Hence, particles with different size and size distribution were obtained through binary blending of the nonpareils while maintaining relatively similar bulk densities. Die fill performance of the nonpareils was unlikely to be affected by differences in bulk densities.

### 3.2. Flow properties of nonpareil blends in relation to particle size

AoR, HR, BFE and SE values for nonpareil blends are shown in Table 2. AoR ranged between 21 and 25° while HR ranged between 1.00 and 1.11. The nonpareils had "excellent" flow properties under compendial AoR and HR classifications (British Pharmacopoeia Commission, 2016) and were considered very free-flowing. It follows that any arching behaviour observed in this study was non-cohesive in nature and would have occurred due to wedging and interlocking of the particles (Schulze, 2007).

BFE ranged between 330 and 610 mJ while SE ranged between 0.54 and 1.07 mJ/g. In general, BFE and SE decreased with increasing particle size ($D_{10}$, $D_{50}$ and $D_{90}$) until a particle size of between 500 and 600 μm, beyond which a plateau effect was observed (Fig. 4).

### 3.3. Reproducibility of results from die filling device

Average nonpareil masses and RSDs across different die orifice diameters are reported in Table 3. With the exception of blend B1 at 2 mm orifice diameter configuration (RSD: 14.25%), all of the RSDs were below 10%, suggesting that results obtained from the die filling device were reproducible and suitable for further analyses. It can also be seen that as orifice diameter decreased from 10 to 2 mm, RSD values generally increased. This could be attributed to the greater difficulty in achieving uniform die fill for smaller orifice diameters.

### 3.4. Arching behaviour under static conditions

Visual observations of arching behaviour were made for nonpareil blends from both Sets A and B across the four orifice diameters under static conditions. Type 0 flow behaviour was observed for 4, 7 and 10 mm diameter orifices for all the nonpareil blends, indicating that no arching behaviour was seen for these orifices. Arching behaviour (Type 1 and Type 2 flow) was only seen for the narrowest 2 mm orifice. Preliminary analyses of particle size and observed flow behaviour revealed that particle size of the largest size fraction, $D_{90}$, was more deterministic for the occurrence of non-cohesive arching. Fig. 5A shows the plot of $D_{90}$ against the classification of orifice flow behaviours for the 2 mm orifice. For particles with $D_{90}$ less than 600 μm, no arching behaviour was observed. However, between $D_{90}$ of 600–800 μm, the nonpareils began to show signs of arching with mixed flow behaviours across Types 0–2. Beyond $D_{90}$ of 800 μm, Type 2 arching became the most prominent behaviour.

### 3.5. Die fill of Set A nonpareil blends in different orifice diameters

Fig. 5B shows the relationship between MFA and $D_{90}$ for Set A nonpareils across 2, 4, 7 and 10 mm diameter die orifices. Relatively good fits of the datasets to second order polynomial functions were achieved ($R^2$: 0.770–0.961). Larger die orifice diameters had higher MFA values since more nonpareil particles were able to enter the orifice per unit time. An increase in $D_{90}$ from 350 to 500 μm was associated with an initial increase in die fill performance of nonpareils across all die orifice diameters with the exception of $MFA_2$, which had instead

**Table 2**
Particle size distribution, bulk densities and flow parameters of nonpareil blends from Sets A and B.

| Code | $D_{10}$ (μm) | $D_{50}$ (μm) | $D_{90}$ (μm) | Span | $\rho_{bulk}$ (g/cm$^3$) | AoR (°) | HR | BFE (mJ) | SE (mJ/g) |
|---|---|---|---|---|---|---|---|---|---|
| A1 | 250.2 | 295.9 | 351.3 | 0.34 | 0.82 | 24.2 (0.1) | 1.09 (0) | 608.9 (17.0) | 1.07 (0.04) |
| A2 | 258.9 | 320.9 | 416.0 | 0.49 | 0.83 | 23.6 (0.5) | 1.08 (0.01) | 450.1 (7.5) | 0.83 (0.01) |
| A3 | 264.5 | 342.0 | 437.5 | 0.51 | 0.85 | 23.2 (0.1) | 1.08 (0) | 392.5 (9.9) | 0.76 (0.01) |
| A4 | 283.5 | 369.1 | 447.5 | 0.44 | 0.85 | 22.9 (0.3) | 1.08 (0) | 391.1 (10.7) | 0.74 (0.02) |
| A5 | 345.0 | 397.8 | 458.1 | 0.28 | 0.86 | 21.8 (0.6) | 1.06 (0) | 422.9 (12.1) | 0.79 (0.01) |
| A6 | 351.2 | 431.4 | 592.6 | 0.56 | 0.86 | 21.9 (0.7) | 1.08 (0.01) | 394.9 (15.3) | 0.78 (0.04) |
| A7 | 360.3 | 486.0 | 634.2 | 0.56 | 0.87 | 23.2 (1.2) | 1.08 (0) | 444.2 (29.9) | 0.77 (0.09) |
| A8 | 374.8 | 530.0 | 647.6 | 0.51 | 0.86 | 23.3 (0.2) | 1.07 (0.01) | 465.8 (5.0) | 0.77 (0.03) |
| A9 | 505.2 | 578.5 | 654.4 | 0.26 | 0.84 | 22.1 (0.1) | 1.06 (0) | 429.9 (1.2) | 0.78 (0.04) |
| A10 | 518.9 | 621.3 | 776.2 | 0.41 | 0.85 | 23.0 (0.1) | 1.06 (0) | 374.2 (44.6) | 0.68 (0.06) |
| A11 | 532.4 | 645.8 | 813.4 | 0.44 | 0.85 | 22.7 (0.4) | 1.06 (0) | 403.0 (54.5) | 0.71 (0.04) |
| A12 | 553.8 | 676.2 | 825.1 | 0.40 | 0.86 | 23.3 (0.7) | 1.06 (0) | 388.6 (18.3) | 0.69 (0.01) |
| A13 | 615.2 | 718.9 | 841.7 | 0.32 | 0.86 | 23.4 (0.3) | 1.06 (0) | 499.8 (6.5) | 0.83 (0.03) |
| B1 | 794.4 | 952.7 | 1077.0 | 0.30 | 0.90 | 24.0 (0.5) | 1.04 (0) | 402.7 (8.7) | 0.80 (0.02) |
| B2 | 189.8 | 919.7 | 1052.3 | 0.94 | 0.94 | 25.0 (0.3) | 1.09 (0) | 379.4 (6.0) | 0.63 (0.03) |
| B3 | 269.9 | 839.9 | 1073.7 | 0.96 | 0.93 | 23.8 (0.4) | 1.09 (0) | 401.7 (4.9) | 0.72 (0.02) |
| B4 | 268.6 | 687.6 | 1069.0 | 1.16 | 0.89 | 24.1 (0.2) | 1.09 (0.01) | 406.3 (6.5) | 0.82 (0.02) |
| B5 | 255.8 | 306.4 | 976.0 | 2.35 | 0.84 | 24.2 (0.1) | 1.09 (0) | 538.1 (11.7) | 0.91 (0.02) |
| B6 | 592.6 | 817.6 | 941.1 | 0.43 | 0.92 | 23.7 (0.7) | 1.06 (0) | 376.9 (4.4) | 0.74 (0.02) |
| B7 | 287.8 | 790.3 | 927.6 | 0.81 | 0.93 | 24.2 (0.5) | 1.09 (0) | 512.4 (7.4) | 0.74 (0) |
| B8 | 284.7 | 778.0 | 923.4 | 0.82 | 0.92 | 24.5 (0.5) | 1.10 (0) | 493.4 (1.6) | 0.79 (0.02) |
| B9 | 262.4 | 350.4 | 913.2 | 1.86 | 0.87 | 24.8 (0.8) | 1.10 (0) | 565.9 (10.4) | 0.88 (0) |
| B10 | 256.4 | 305.6 | 832.1 | 1.88 | 0.83 | 23.7 (0.6) | 1.08 (0.02) | 412.4 (6.7) | 0.83 (0.06) |
| B11 | 496.1 | 702.9 | 824.3 | 0.47 | 0.88 | 23.3 (0.4) | 1.06 (0) | 492.7 (7.1) | 0.70 (0.02) |
| B12 | 291.5 | 688.6 | 824.3 | 0.77 | 0.91 | 24.1 (0.8) | 1.08 (0.02) | 413.3 (7.3) | 0.61 (0.02) |
| B13 | 280.2 | 676.2 | 828.9 | 0.81 | 0.89 | 23.7 (0.8) | 1.11 (0) | 423.5 (9.9) | 0.71 (0.01) |
| B14 | 262.6 | 342.5 | 793.4 | 1.55 | 0.87 | 23.9 (0.8) | 1.10 (0) | 488.4 (3.1) | 0.82 (0.02) |
| B15 | 258.7 | 307.1 | 715.8 | 1.49 | 0.83 | 22.6 (0.7) | 1.09 (0) | 417.1 (0.5) | 0.79 (0.02) |
| B16 | 496.4 | 574.2 | 649.0 | 0.27 | 0.86 | 21.9 (0.6) | 1.08 (0) | 331.6 (8.9) | 0.54 (0.01) |
| B17 | 311.5 | 563.9 | 646.5 | 0.59 | 0.87 | 23.6 (0.3) | 1.09 (0.01) | 483.2 (51.8) | 0.68 (0.05) |
| B18 | 271.1 | 517.6 | 633.6 | 0.70 | 0.86 | 22.8 (0.8) | 1.10 (0) | 422.8 (6.9) | 0.71 (0.03) |
| B19 | 260.7 | 325.7 | 619.6 | 1.10 | 0.85 | 23.2 (0.4) | 1.10 (0) | 457.1 (3.3) | 0.79 (0.03) |
| B20 | 252.7 | 301.0 | 400.6 | 0.49 | 0.82 | 22.2 (0.5) | 1.09 (0.01) | 364.6 (4.5) | 0.88 (0.01) |

Parentheses indicate the respective standard deviations.

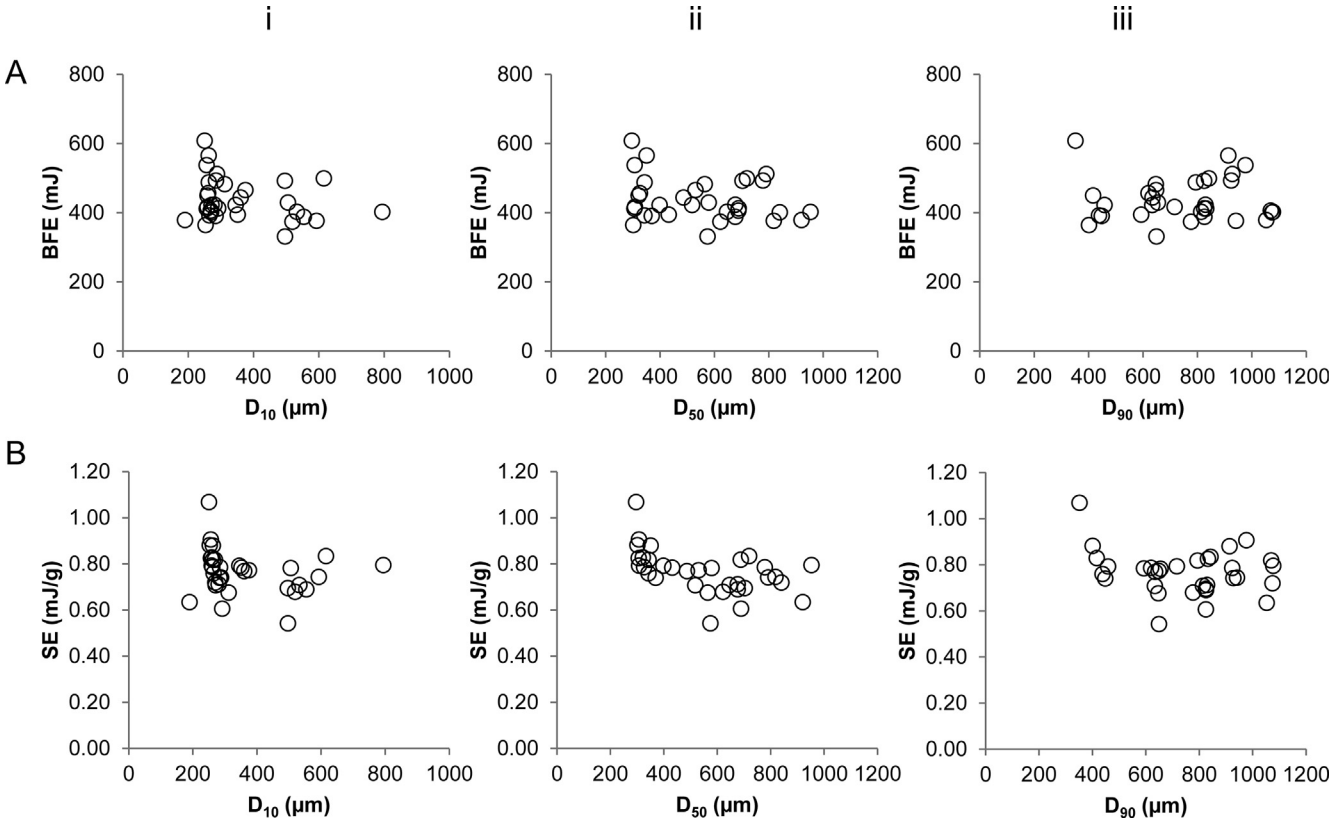



**Fig. 4.** Relationship between particle size parameters (i) $D_{10}$, (ii) $D_{50}$ and (iii) $D_{90}$, and flow parameters (A) BFE and (B) SE for all nonpareil blends.

**Table 3**
Average collected nonpareil masses and RSDs across different die orifice diameters.

| Code | Orifice diameter (mm) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 10 | | 7 | | 4 | | 2 | |
| | Average mass (g) | RSD (%) | Average mass (g) | RSD (%) | Average mass (g) | RSD (%) | Average mass (g) | RSD (%) |
| A1 | 3.20 | 4.49 | 1.36 | 2.21 | 0.420 | 0.49 | 0.071 | 1.79 |
| A2 | 3.47 | 3.53 | 1.47 | 0.54 | 0.469 | 0.62 | 0.069 | 2.30 |
| A3 | 3.51 | 2.81 | 1.52 | 0.48 | 0.476 | 3.72 | 0.069 | 2.82 |
| A4 | 3.64 | 3.48 | 1.56 | 0.57 | 0.467 | 2.34 | 0.068 | 2.20 |
| A5 | 4.02 | 2.00 | 1.66 | 1.14 | 0.480 | 1.44 | 0.069 | 1.76 |
| A6 | 4.32 | 2.32 | 1.66 | 1.65 | 0.476 | 2.75 | 0.063 | 2.45 |
| A7 | 4.40 | 1.10 | 1.68 | 0.12 | 0.464 | 0.98 | 0.058 | 4.54 |
| A8 | 4.39 | 0.56 | 1.63 | 0.04 | 0.455 | 1.56 | 0.058 | 2.54 |
| A9 | 4.46 | 1.50 | 1.60 | 0.62 | 0.432 | 1.87 | 0.049 | 4.26 |
| A10 | 4.47 | 0.57 | 1.59 | 0.50 | 0.419 | 1.98 | 0.071 | 3.43 |
| A11 | 4.46 | 0.93 | 1.57 | 0.46 | 0.421 | 1.39 | 0.071 | 3.64 |
| A12 | 4.52 | 1.55 | 1.60 | 1.18 | 0.413 | 1.39 | 0.071 | 5.02 |
| A13 | 4.46 | 2.32 | 1.55 | 0.54 | 0.401 | 4.28 | 0.071 | 4.28 |
| B1 | – | – | – | – | – | – | 1.95 | 14.25 |
| B2 | – | – | – | – | – | – | 2.63 | 9.89 |
| B3 | – | – | – | – | – | – | 3.05 | 5.09 |
| B4 | – | – | – | – | – | – | 3.30 | 3.88 |
| B5 | – | – | – | – | – | – | 3.58 | 3.01 |
| B6 | – | – | – | – | – | – | 2.21 | 9.39 |
| B7 | – | – | – | – | – | – | 2.84 | 5.31 |
| B8 | – | – | – | – | – | – | 3.02 | 5.04 |
| B9 | – | – | – | – | – | – | 3.30 | 3.24 |
| B10 | – | – | – | – | – | – | 3.65 | 1.84 |
| B11 | – | – | – | – | – | – | 2.34 | 5.38 |
| B12 | – | – | – | – | – | – | 2.84 | 6.03 |
| B13 | – | – | – | – | – | – | 3.01 | 6.16 |
| B14 | – | – | – | – | – | – | 3.27 | 3.15 |
| B15 | – | – | – | – | – | – | 3.51 | 2.02 |
| B16 | – | – | – | – | – | – | 2.74 | 4.70 |
| B17 | – | – | – | – | – | – | 2.98 | 5.60 |
| B18 | – | – | – | – | – | – | 3.17 | 2.87 |
| B19 | – | – | – | – | – | – | 3.42 | 3.02 |
| B20 | – | – | – | – | – | – | 3.50 | 2.11 |

showed a very slight decrease in die fill for 2 mm diameter orifice. This particle size range largely coincided with that seen for the decrease in SE (Fig. 4B(iii)). Beyond 500–600 μm, die fill performance of the nonpareils generally started to decrease as seen from the decline in $MFA_2$, $MFA_4$ and $MFA_7$ while a plateau effect was seen for $MFA_{10}$. The extent of decrease in MFA was greater for the smaller orifice diameters. In order to determine if there was a critical ratio between the orifice diameter and $D_{90}$ ($R_{orifice/D90}$) on die fill performance, MFA values for all of the orifice diameters were scaled from a range of 0–1 and plotted against $R_{orifice/D90}$ as shown in Fig. 5C. Second order polynomial functions were used to fit the datasets ($R^2$: 0.821–0.921). The approximate critical $R_{orifice/D90}$ values to obtain maximum MFA in the different orifice diameters were as follows; $MFA_2$: 5, $MFA_4$: 8, $MFA_7$: 12 and $MFA_{10}$: 14. While a particle size of 500–600 μm appeared to be the critical size range for die fill performance across the different orifice diameters, a common critical $R_{orifice/D90}$ value could not be identified.

### 3.6. Die fill of Set B nonpareil blends in 2 mm orifice diameter

Decrease in die fill of Set A nonpareil blends with increase in particle size was most apparent for 2 mm orifice diameter which suggested that non-cohesive arching had greatest influence over die fill in the narrowest orifice. Set A nonpareil blends had relatively narrow Span values ranging between 0.3 and 0.6 (Table 2). In order to investigate how particle size distribution could potentially influence arching behaviour and die fill performance in the 2 mm orifice, Set B nonpareils were prepared from binary blends of non-neighbouring size fractions to obtain particles with wider size distributions. It was also of interest to study how the effect of forced feeding, simulated by paddle inserts with different angles of inclination, could affect arching behaviour and die fill performance in such a narrow orifice diameter.

Fig. 6A shows $MFA_2$ against $D_{90}$ for nonpareil blends from Sets A and B at a paddle angle configuration of 90°. For Set B nonpareils, it could be seen that beyond $D_{90}$ value of 500–600 μm, $MFA_2$ started to show a more dispersed behaviour albeit the general trend was a decrease in $MFA_2$ as $D_{90}$ increased. This was largely in line with observations made for Set A nonpareils. Set B nonpareils that had both large particle size and wide size distribution, as indicated by large $D_{90}$ and span values, had relatively high $MFA_2$ values as well, which might have accounted for the spread in data beyond $D_{90}$ value of 500–600 μm. Surface plots of $D_{90}$, Span and $MFA_2$ (Fig. 6B) showed that the highest $MFA_2$ values were associated with low values of $D_{90}$ but large Span values. On the other hand, nonpareils with larger $D_{90}$ and low Span values had the lowest $MFA_2$ values. It follows that both $D_{90}$ and Span affected $MFA_2$ values.

Fig. 6C shows the plots of $MFA_2$ against $D_{90}$ for different paddle angles across all the nonpareil blends from Sets A and B. All three paddle angles showed very similar die fill performance with the same general features: a plateau area between $D_{90}$ of 350–500 μm followed by a general decrease and scattering in $MFA_2$ values beyond 500–600 μm. It follows that the different paddle angles did not affect the relationship between $D_{90}$ and die fill performance.

## 4. Discussion

### 4.1. Sensitivity of flow measurements

Powder flow is affected by a complex interplay of particle

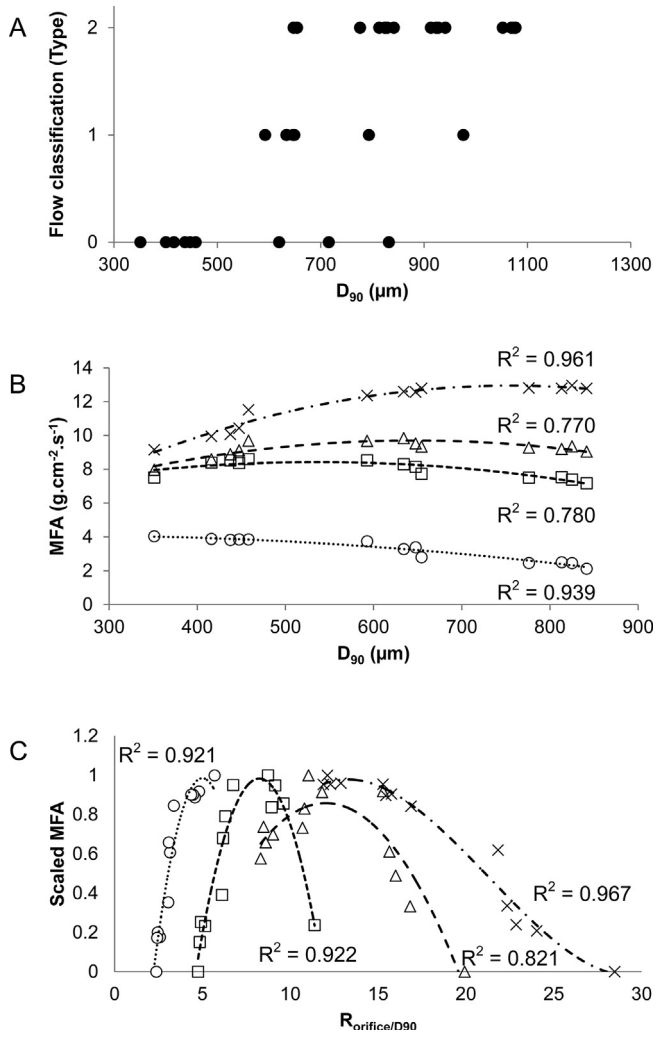



**Fig. 5.** (A) Plot of orifice flow classification against $D_{90}$ for 2 mm orifice. (B) Plots of MFA against $D_{90}$ for different orifice diameters. (C) Relationship between scaled MFA and $R_{orifice/D90}$. MFA: $MFA_2$ (○), $MFA_4$ (□), $MFA_7$ (△) and $MFA_{10}$ (×).

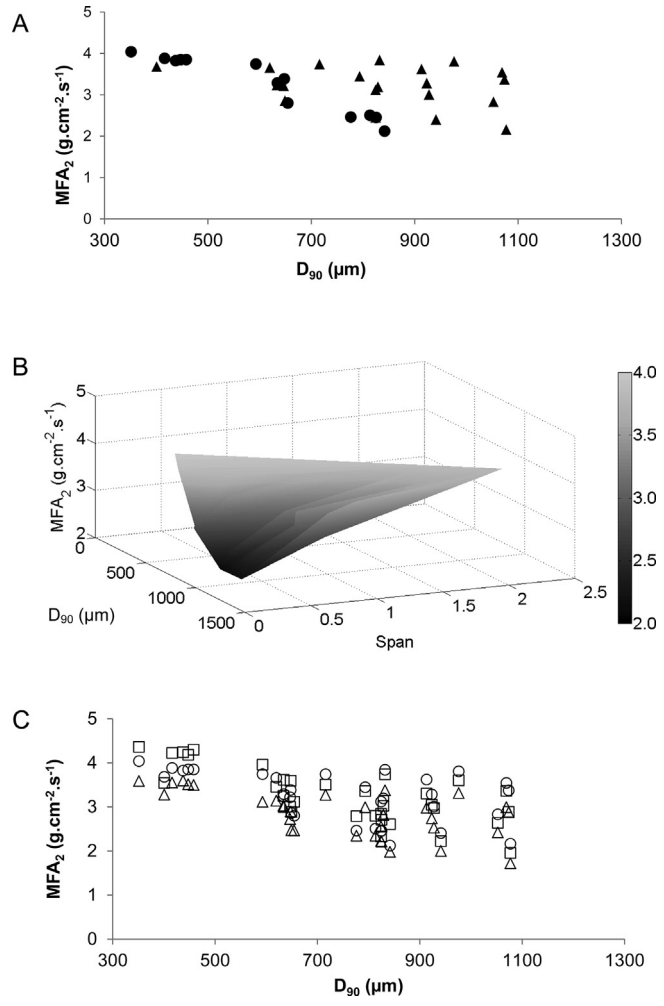



**Fig. 6.** (A) Plot of $MFA_2$ against $D_{90}$ for nonpareil blends from Set A (●) and Set B (▲) at paddle angle of 90°. (B) Surface plot of $D_{90}$ and Span against $MFA_2$ for paddle angle of 90°. (C) Plot of $MFA_2$ against $D_{90}$ for different paddle angles. Paddle angles: 30° (△), 60° (□), and 90° (○).

properties, such as particle size and shape, surface interactions and external stresses. To date, there is no single flow characterisation technique that can adequately account for all these factors. A combination of multiple standardised methods is often used to describe powder flow behaviour (Lee et al., 2000; Sandler and Wilson, 2010). In this study, the flow properties of the nonpareil blends were characterised by AoR, HR, BFE and SE. Both AoR and HR values had narrow ranges of values across both Sets A and B nonpareils blends (AoR: 21–25°, HR: 1.07–1.09) and ranked the flow for all of the nonpareil blends as "excellent" under compendial classification. This might indicate that both AoR and HR were likely to be measuring or affected by similar physical characteristics of the blends. Consequently, it was difficult to discriminate the flow properties across the blends using just AoR and HR.

Newer methods of powder flow characterisation, such as powder rheometry, is potentially advantageous at differentiating flow of non-cohesive materials (Osorio and Muzzio, 2013). For non-cohesive and coarse particles, such as the nonpareil blends used in this study, the interaction between the particles is mainly influenced by steric repulsion and friction forces (Lumay et al., 2012). During the powder rheometry tests, the energy expended by the twisted blade in overcoming the inter-particulate interactions was recorded as either BFE or SE, depending on the blade's direction of motion.

An increase in particle size was expected to reduce BFE and SE due to reduction in inter-particulate contact area and friction. While a general decrease in SE with particle size was seen (Fig. 4B(i)–B(iii)), the correlation between BFE and particle size was less apparent (Fig. 4A(i)–A(iii)). BFE is measured under high stress flow conditions and is inherently a multidimensional flow parameter which considers shear strength, cohesion, compressibility, inter-particulate friction amongst other physical properties of the sample (Freeman, 2007; Leturia et al., 2014). HR, an indicator of powder compressibility, was comparable across the nonpareil blends (Table 2). Poor correlation between BFE with particle size could have resulted from non-specificity to inter-particulate friction. In contrast, SE has a higher specificity for inter-particulate friction (Tay et al., 2017; Zhou et al., 2010) possibly due to lower compressive forces generated during the impeller blade's upward traverse through the sample. This could account for the better correlation of SE with particle size.

### 4.2. Effect of particle size and size distribution on arching and die fill

Powder flow behaviour in hoppers and silos can be considered as being static since the particles are initially at rest before they exit through the orifice. Parameters such as avalanche size distribution have been commonly used to describe arching events under such static conditions. In this study, the arching behaviour under static flow conditions was simplified as being either Types 0, 1 or 2. For the case of die

filling, the particles were already in motion where they had to traverse across the orifice before entering it. Consequently, the use of parameters which describe arching under static flow conditions may no longer be suitable. Die fill performance, characterised by MFA in this study, was used as an indicator of arching during die fill under gravity.

All of the nonpareil blends prepared in this study were relatively coarse and round in shape with excellent flow properties, implying that cohesive arching was highly unlikely. Moreover, inter-particulate friction decreased when particle size increased (Fig. 4B(i)–B(iii)). It follows that die fill performance or MFA should have increased with particle size. However, this was only observed for $MFA_{10}$ (Fig. 5B). For the 2–7 mm orifices, increasing the particle size beyond 500–600 μm appeared to have a detrimental effect on die fill. The most likely reason for this decrease in die fill with increase in particle size was non-cohesive arching where the particles were simply too large, resulting in interlocking and jamming which in turn limited entry of particles into the orifice. For the 10 mm orifice, no decrease in die fill was observed which suggested that no arching was possible for the particle size of nonpareils used in the study. Interestingly, the increase in die fill for the 10 mm orifice with particle size coincided with the decrease in SE, suggesting that the die fill was primarily affected by inter-particulate friction.

Frictional forces with the orifice wall could also have affected the powder flow behaviour (Kachrimanis et al., 2005). The smaller the die orifice, the greater the wall effects on powder flow since a larger portion of the flow occurs in close proximity to the wall, creating drag forces which can slow down the particle movement. However, it is unlikely that wall effects contributed significantly to the powder flow behaviour since the depth of the orifice was only 20 mm. Moreover, larger particles had poorer die fill performance despite lesser surface area for frictional contact with the die wall. These suggest that non-cohesive arching, rather than wall effects, had the greatest impact on powder flow behaviour.

It was worth noting that no arching event was recorded for 4 and 7 mm orifices under static conditions albeit the MFA data showed otherwise. This highlights an important difference in arching under static and dynamic flow conditions. Under conditions simulated on the die fill device, the particles accelerated from rest and approached the orifice at a certain velocity (in this case 4 m/min). As they moved across the orifice, the particles had a tendency to continue in their line of motion as a result of momentum instead of flowing though the orifice immediately, more so for larger particles which were heavier in mass. Moreover, due to the speed of approach, the apparent orifice diameter would have also become smaller with respect to the moving particles. These increase the likelihood of arching under dynamic, die fill conditions, which could not be captured under static conditions. Increasing the speed of the particles' movement across the orifice might be expected to increase the arching effect and further decrease die fill performance under gravity.

While it was apparent that the likelihood of non-cohesive arching increased with increasing particle size (Fig. 5A), particle size distribution had also affected arching behaviour. Particles with large $D_{90}$ values could still achieve good die fill performance in the narrow 2 mm orifice provided that their Span values were also high (Fig. 6A and B). On the other hand, $MFA_2$ was the lowest for particles with large $D_{90}$ and small Span values. This might be attributed to the probability of large particles interlocking at the orifice; a wide size distribution or particles with small $D_{90}$ values reduce the likelihood of large particles meeting at the orifice and hence arching at the orifice.

A common $R_{orifice/D90}$ across all the orifice diameters to attain maximum die fill without the influence of non-cohesive arching could not be clearly identified in this study (Fig. 5C). Nonetheless, a critical $D_{90}$ size range of 500–600 μm was found for the materials investigated (Fig. 5B). Process conditions, such as paddle speed, and particle physical characteristics, such as particle shape, are also likely to affect this critical particle size range. Increasing $D_{90}$ beyond 600 μm had greater impact for $MFA_2$ and $MFA_4$ than $MFA_7$ as the negative gradients of the fitted curves were much steeper in comparison and could have implications for certain applications where narrow orifices are encountered as in mini-tablet production. Typically, high die fill performance is desired since higher filling densities can be achieved which in turn reduces compression problems (Reed, 1995). It follows that non-cohesive arching during die fill is more likely to have greater impact for narrow orifice diameters and a $D_{90}$ size range not exceeding 500–600 μm could be useful as an early reference for particle size specifications to avoid this problem.

### 4.3. Effect of forced feeding on arching and die fill

For most rotary tablet presses, the tablet feed is circulated in a feed frame and forced fed into the die orifice by action of rotating paddles. In this study, paddle inserts with different angles of inclination were used to simulate different extents of forced feeding. The more acute the angle, the greater the forced feeding effect expected since the downward component of the force towards the die orifice would increase. However, the die fill performance across the different paddle angles appeared to be similar across the different paddle angles (Fig. 6C). Paddles with small angles of inclination have greater surface area of contact with the particles (30°–538 $mm^2$, 60°–337 $mm^2$ and 90°–300 $mm^2$). Consequently, the overall force from the feeder is dissipated across a larger number of particles for smaller paddle angles despite a greater component of downward force present. This could possibly account for the similar die fill performance across the different paddle angles. The different paddle angles also did not appear to affect the $D_{90}$ critical range, suggesting that the incidence of arching during die fill was not affected by forced feeding but dependent on other process parameters or material inherent characteristics, in particular, particle size.

## 5. Conclusion

In this study, non-cohesive arching of coarse and free-flowing nonpareils during die fill by gravity was investigated using a special die fill device. A methodology based on die fill performance was used to evaluate non-cohesive arching during die filling. It was demonstrated that die fill performance could better reflect the dynamic conditions and hence capture the arching phenomenon during die fill than observations made under static flow conditions. This was attributed to the particles' momentum and reduced apparent orifice diameter as they had accelerated from rest and approached the orifice with a certain velocity. For narrow orifice diameters, die fill performance decreased with increasing particle size due to non-cohesive arching. In contrast, die fill performance increased with increasing particle size for the larger orifice diameters in general due to a decrease in inter-particulate friction. Both particle size and size distribution were found to affect the likelihood of non-cohesive arching under die fill conditions. Forced feeding did not appear to have significant effect on die fill performance and non-cohesive arching for the 2 mm orifice. It was also shown that non-cohesive arching was more likely to have great impact for narrow orifices which could potentially be mitigated by using particles below the critical particle size range identified in this study. It should however be noted that process conditions and other particle properties such as shape could influence the critical size range. Findings from this study might be useful for addressing problems associated with non-cohesive arching under dynamic flow conditions and potentially provide some guidance on particle size specifications for certain pharmaceutical applications such as mini-tablet production.

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